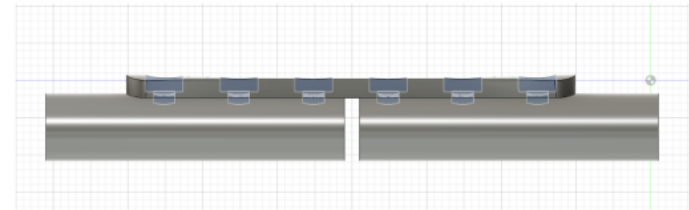


Project 1: Finite Element Analysis of a Femoral Bone Plate

Stress Shielding & Fracture Fixation Analysis — Autodesk Fusion 360

Overview

A six-screw femoral bone plate was modeled in Autodesk Fusion 360 and applied across a simulated 2 mm fracture gap in a hollow chicken femur, mimicking a clinical fracture-fixation scenario. Three screws were placed on each side of the fracture to secure the plate to the intact bone segments. The model was used to evaluate **stress shielding**, the phenomenon in which an overly stiff plate carries a disproportionate share of the mechanical load, reducing stress transfer to the healing bone and risking **aseptic loosening** and delayed union.



Figure

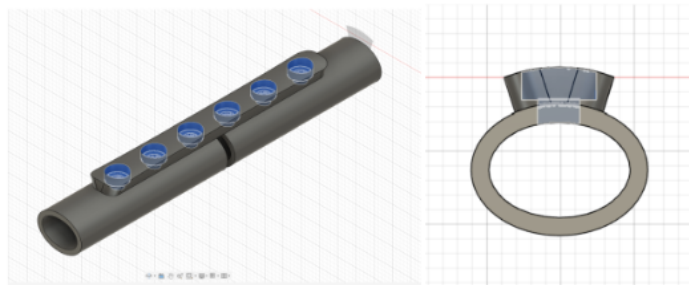


Figure 1: CAD model of the six-screw bone plate spanning the fracture gap.

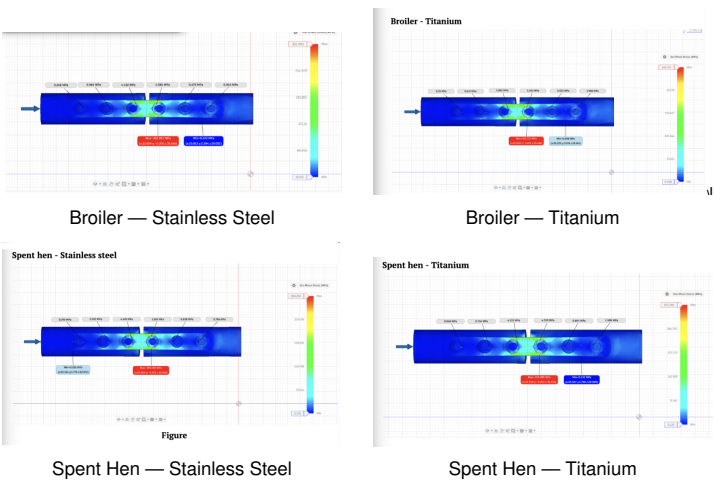
Methodology

- Modeled a femur plate on a hollow chicken bone (**broiler** vs. **spent hen**), each previously characterized in a theoretical fracture analysis.
- Applied an axial load equal to 250% of estimated body weight (49.0 N broiler, 24.5 N spent hen) to represent dynamic loading.
- Ran static stress simulations comparing **stainless steel** and **titanium** plates for each bone type.
- Evaluated results using the Von Mises stress distribution, where red denotes peak stress concentration, blue denotes stress shielding (load bypassing the bone), and green/yellow denotes sufficient, sub-failure load transfer.

Conclusion

The stress shielding ratio (SSR) was consistently lower for titanium plates in both bone models, indicating that titanium carries a comparatively larger share of the load relative to the bone, which theoretically raises the risk of aseptic loosening. However, titanium's lower Young's modulus more closely matches that of bone, allowing the plate to transfer load back to the healing tissue and reduce the stiffness mismatch that drives stress shielding. Titanium also

FEA Results: Von Mises Stress



Summary of Results

Chicken	Body mass (kg)	Body weight (N) $W = m \cdot g$	Estimated Load (N) $L = W \cdot 250\%$
Broiler chicken	2	19.6	49.0
Spent hens	1	9.81	24.5

Chicken	Plate material	Max Von Mises stress in the plate (MPa)	Safety Factor $\frac{\sigma_y}{\sigma_{max}}$	Average stress in the bone under the plate (MPa)	Average stress in the intact bone (MPa)	SSR (%) $\frac{\sigma_{intact} - \sigma_{plated}}{\sigma_{intact}}$
Broiler chicken	Stainless steel*	302.993	0.561	~ 1.8910	~ 6.6538	71.580
	Titanium**	265.713	3.321	~ 2.4003	~ 6.6538	63.925
Spent hens	Stainless steel*	399.259	0.426	~ 1.8022	~ 8.8763	79.700
	Titanium**	355.285	2.484	~ 2.1047	~ 8.8763	76.289

*Stainless stress yield strength: 170.00 MPa

** Titanium yield strength: 882.528 MPa

produced consistently lower peak Von Mises stress and a higher safety factor across both chicken models, indicating more efficient load distribution across the plate-bone system and a lower likelihood of plate failure under physiological loading. In both materials, the maximum stress concentrated at the screw hole closest to the fracture gap, identifying it as the critical failure-initiation site in this fixation design.

Project 2: Screw Fixation Animation & Femoral Impingement Test

Ball-and-Socket Joint Range-of-Motion Analysis — Autodesk Fusion 360

Overview

Building on the Project 1 bone plate model, this study animated the physical installation of the fixation hardware and then evaluated how the installed plate affects joint range of motion. Socket head cap screws were driven through the plate and into the fractured femur segments using a Fusion 360 joint-motion animation, replicating the surgical sequence of locking the plate against the bone. Once fixation was complete, the femoral head was rotated within its socket to identify the flexion angle at which the plate first causes mechanical **impingement** against the socket rim — a key constraint in surgical planning for plate length and placement.

Milestone:

- Screws Inserted on Rigid Setup:



Start of Animation (Screws & Plate Detached)

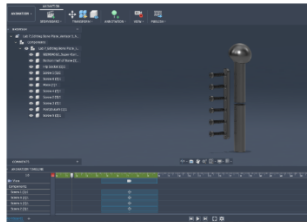


Figure 2: Rigid screw setup (top) and start of the fixation animation with screws and plate detached (bottom).

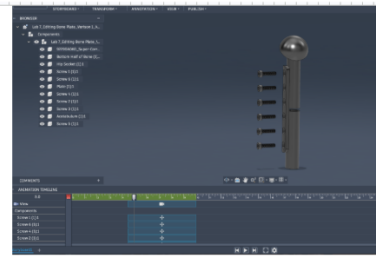
Methodology

- Modeled six socket head cap screws and animated their insertion through the plate and into both fractured bone segments, locking the plate in a fixed position.
- Used Fusion 360's joint and animation timeline tools to drive the screws axially into the bone, confirming clash-free insertion at each stage.
- Applied a ball-and-socket joint at the femoral head to simulate flexion of the hip joint with the plate rigidly fixed in place.
- Incrementally rotated the femoral head within the socket and visually inspected each angle for contact/impingement between the plate and the socket rim.
- Repeated the rotation sweep using a 60 mm plate to determine the maximum achievable flexion before impingement occurred.

Conclusion

The animated fixation sequence confirmed that the six-screw plate could be installed on the fractured femur without geometric clashes between the screws, plate, and bone. The subsequent rotation sweep showed that the femoral head could reach **105° of flexion** before the 60 mm plate impinged on the socket rim, establishing

Fixation Animation Sequence



Near-End of Animation (Rotation Test)

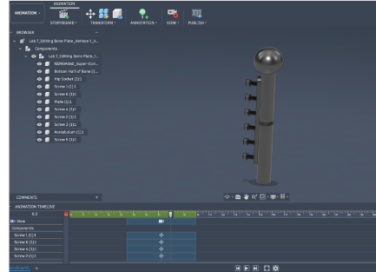
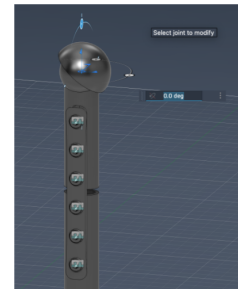


Figure 3: Mid-animation screw insertion (top) and near-end rotation test with screws locked (bottom).

Flexion / Impingement Test

2. Report the maximum flexion achieved when rotating the femur and plate.

0° Rotation:



Max Flexion: 45° Rotation

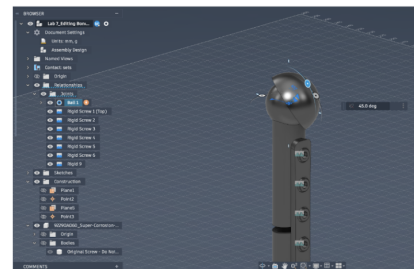


Figure 4: 0° baseline rotation (top) vs. 45° flexion during the joint rotation sweep (bottom).

105°

Maximum flexion achieved before impingement (60 mm plate)

a practical range-of-motion limit imposed by plate length and position. This result highlights a key design trade-off in fracture fixation: while longer plates improve fixation stability and load distribution (as shown in Project 1), they can reduce the achievable joint range of motion, so plate length and screw placement must be balanced against the patient's expected post-operative mobility needs.

Project 3: Frog Gastrocnemius Muscle Mechanics

Length-Tension & Force-Velocity Relationships — Electrical Stimulation Study

Overview

The frog gastrocnemius muscle was studied under electrical stimulation on a force transducer platform, formally characterizing two classic skeletal muscle properties: the **length-tension** and **force-velocity** relationships. Isometric twitch responses were first used to establish stimulation parameters, after which the muscle was systematically stretched and loaded to generate each relationship.

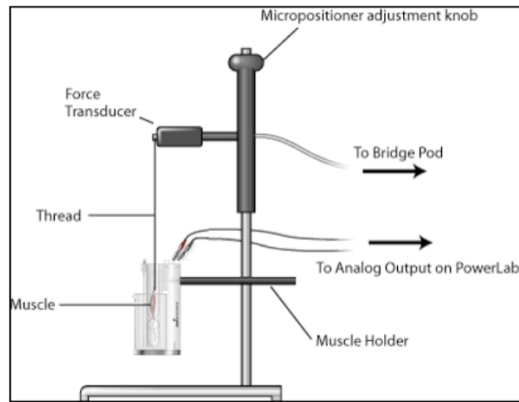
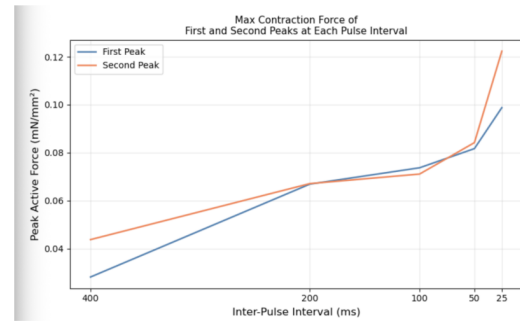


Figure 5: Force transducer platform: muscle held in Ringer's solution, connected via thread to the force transducer and PowerLab acquisition system.

Methodology

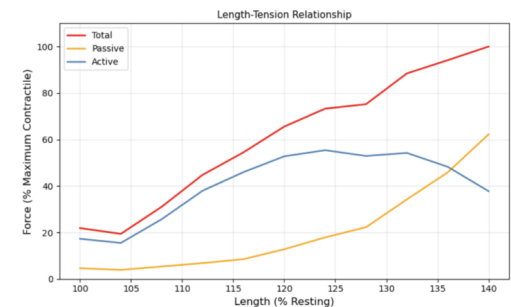
- Mounted the excised gastrocnemius in a muscle holder, connecting it via thread to a force transducer feeding a PowerLab bridge amplifier.
- Delivered isometric twitch stimuli at a fixed pulse width to determine the maximum excitation voltage and the muscle's supramaximal stimulus voltage.
- Applied paired stimulus pulses across decreasing inter-pulse intervals (400 ms to 25 ms), recording first- and second-peak active force to evaluate twitch summation.
- Incrementally stretched the muscle from 100% to 140% of resting length (L_0), recording total, passive, and active force at each length.
- Measured shortening velocity under a range of applied loads and fit the results in MATLAB to Hill's hyperbolic equation, $v = a(V_{max} - F)/(F + b)$, via bounded nonlinear regression.

Twitch Summation



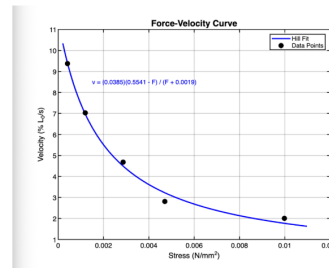
Peak active force rose modestly as inter-pulse interval decreased from 400 ms to 100 ms, then increased sharply below 50 ms, marking the transition from partial to near-complete summation as stimulation frequency approached the fusion frequency.

Length-Tension Relationship

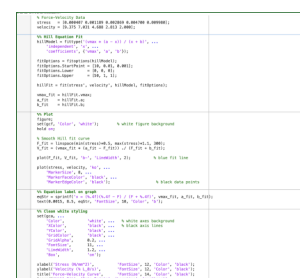


Active force peaked near 120–132% L_0 before declining on the descending limb, consistent with reduced actin–myosin overlap at longer sarcomere lengths; passive force rose sharply beyond 130% L_0 as elastic connective tissue engaged.

Force-Velocity Relationship



Hill Fit Curve



MATLAB Code

Shortening velocity decreased hyperbolically with increasing load, matching Hill's characteristic equation with fitted constants $V_{max} \approx 0.0385$, $a \approx 0.554$, $b \approx 0.0019$.

Conclusion

Together, the three relationships define the physiological operating envelope of skeletal muscle contraction. The twitch summation data show how repeated stimulation temporally recruits additional contractile force as pulses fuse, the length-tension curve confirms the sliding-filament theory of overlap-dependent force generation

with peak active tension at mid-range sarcomere length, and the force-velocity curve validates Hill's model for the inverse trade-off between load and shortening speed. These findings collectively reinforce the mechanistic basis for how stimulation frequency, muscle length, and applied load interact to determine contractile output.

Project 4: Arduino Smart Bandage for Wound Monitoring

Real-Time Temperature & Humidity Alert System

Overview

To address improper wound care monitoring, a smart bandage prototype was designed using an Arduino Uno paired with a DHT11 temperature and humidity sensor to continuously track wound-site conditions for early signs of excessive fluid buildup or infection. Sensor readings were processed on the microcontroller and used to trigger a visual LED alert whenever conditions moved outside a defined healthy range, giving caregivers an immediate signal that a dressing may need attention.

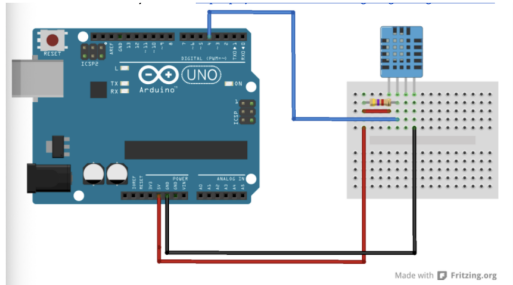


Figure 6: DHT11 sensor wired to an Arduino Uno on a breadboard, powered from the 5V rail with signal read on a digital pin.

Methodology

- Wired a DHT11 temperature/humidity sensor to an Arduino Uno on a breadboard, powering it from the 5V rail and reading data on a digital pin.
- Set healthy baseline ranges of 32–36°C and 40–60% relative humidity to reflect typical wound-surface conditions.
- Flagged an alert for a sudden rise of 1–2°C or readings consistently above 37–38°C, and for humidity exceeding 70–80%, both markers of inflammation or excess exudate.
- Polled the sensor every 500ms and streamed temperature/humidity readings to the serial monitor to validate sensor accuracy before adding alert logic.
- Extended the base firmware with threshold logic driving a digital output pin, lighting a blinking red LED whenever either threshold was exceeded.
- Sized and packaged the sensor to sit flush inside a prototype bandage without interfering with wound contact or patient comfort.

Conclusion

The completed prototype demonstrates a low-cost, real-time wound monitoring system capable of flagging early signs of infection or excess fluid buildup without requiring constant caregiver inspection of the dressing. Because the alert logic is threshold-configurable in software, the system can be tuned to different wound types or pa-

Sensor Reading Code

ACTUAL CODE FOR SENSOR READINGS ALONE:

```
#include <Adafruit_AHTX0.h>

Adafruit_AHTX0 aht;

void setup() {
  Serial.begin(115200);
  Serial.println("Adafruit AHT10/AHT20 demo!");

  if (!aht.begin()) {
    Serial.println("Could not find AHT? Check wiring");
    while (1) delay(10);
  }
  Serial.println("AHT10 or AHT20 found");
}

void loop() {
  sensors_event_t humidity, temp;
  aht.getEvent(&humidity, &temp); // populate temp and humidity objects with fresh data
  Serial.print("Temperature: "); Serial.print(temp.temperature); Serial.println("
degrees C");
  Serial.print("Humidity: "); Serial.print(humidity.relative_humidity);
  Serial.println("% rH");

  delay(500);
}
```

Baseline firmware initializes the AHT sensor and streams temperature and humidity readings to the serial monitor every 500 ms, used to validate sensor accuracy before adding alert logic.

Threshold & LED Alert Logic

```
#include <Adafruit_AHTX0.h>

Adafruit_AHTX0 aht;

const int ledPin = 8; // RESTING VALUES 20 DEG C and 30 HUMID
const float tempThreshold = 37.0; // Temperature threshold in °C (Device: 37 Demo:
25/30)
const float humidityThreshold = 75.0; // Humidity threshold in % (Device: 75% Demo:
55%)

void setup() {
  Serial.begin(115200);
  pinMode(ledPin, OUTPUT);

  Serial.println("Adafruit AHT10/AHT20 demo!");

  if (!aht.begin()) {
    Serial.println("Could not find AHT? Check wiring");
    while (1) delay(10);
  }
  Serial.println("AHT10 or AHT20 found");
}

void loop() {
  sensors_event_t humidity, temp;
  aht.getEvent(&humidity, &temp); // populate temp and humidity objects with fresh data

  Serial.print("Temperature: "); Serial.print(temp.temperature); Serial.println("
degrees C");
  Serial.print("Humidity: "); Serial.print(humidity.relative_humidity);
  Serial.println("% rH");

  // LED logic: turn on if temperature OR humidity exceeds threshold
  if (temp.temperature >= tempThreshold || humidity.relative_humidity >=
humidityThreshold) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }

  delay(500);
}
```

Extended logic compares each reading against the temperature and humidity thresholds and drives `ledPin` HIGH to trigger the alert whenever either condition is exceeded.

tient risk profiles, and pairing an absolute threshold with a sudden-change threshold allows it to catch both slow, chronic complications and acute changes in wound status. Future iterations could add wireless reporting so caregivers are notified remotely rather than relying on a visible LED alone.

Project 5: 3D Modeling of a Prototype EEG Cap

Anatomically Conforming Electrode Placement and CAD Development

Overview

As part of my EEG research, I began developing a 3D CAD prototype of an EEG cap designed to conform to the geometry of a human head. The project combines anatomical modeling, standardized electrode placement, and CAD development to create a foundation for a functional EEG-cap prototype.

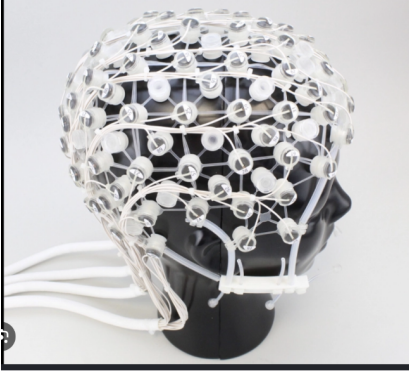


Figure 7: Commercial EEG cap used as the reference for the prototype.

Methodology

- Created a 3D human-head model using Blender and transferred the geometry into Autodesk Fusion.
- Used the standardized 10–20 EEG system to establish electrode locations.
- Simplified the full electrode array to six representative locations for initial CAD development.
- Created reference lines across the head surface to define the electrode grid.
- Planned to conform the grid to the head surface before adding the remaining electrodes and cap structure.

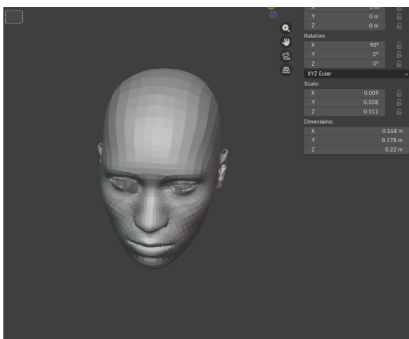


Figure 8: Human-head geometry modeled in Blender.

Electrode Grid Development

The initial electrode layout was reduced to six locations to simplify the first stage of the 3D modeling process. This provides a manageable framework for establishing the grid geometry before expanding to the complete electrode array.

Electrode Placement

The electrode layout was based on the standardized 10–20 EEG positioning system. Electrode locations are defined relative to anatomical landmarks on the head, providing a consistent framework for transferring the layout into the 3D CAD environment.

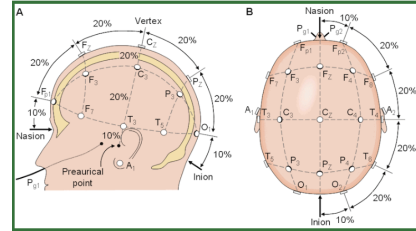


Figure 9: Standardized 10–20 electrode-placement system used as the design reference.

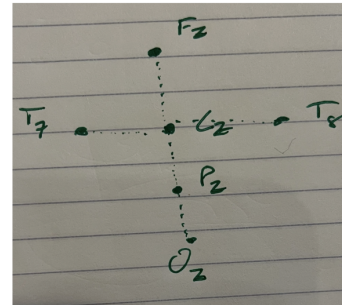


Figure 10: Initial six-electrode grid concept.

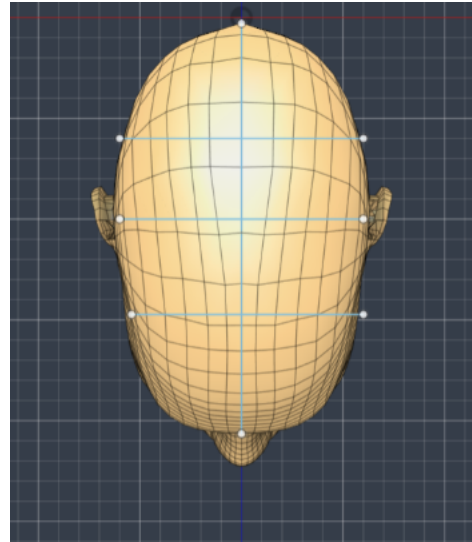


Figure 11: Electrode reference points mapped onto the 3D head in Autodesk Fusion.

CAD Development

The six-electrode layout was transferred into Autodesk Fusion and positioned relative to the head's anatomical landmarks. Reference curves were drawn across the head surface to establish the electrode locations and create the initial grid. The next stage is to conform these reference layers to the curved surface and use them as the foundation for the cap's supporting structure.

Conclusion

The current model establishes the initial CAD framework for an anatomically conforming EEG cap. By combining a 3D head model with standardized electrode coordinates, the electrode grid can be developed directly on the curved head surface rather than as a flat layout. The next stage will be to wrap the grid onto the head, incorporate the remaining electrodes, and develop the supporting cap structure and materials.